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Original Contribution

# Optimal blood pressure decreases acute kidney injury after gastrointestinal surgery in elderly hypertensive patients: A randomized study☆,☆☆,☆☆☆,☆☆☆☆

# Optimal blood pressure reduces acute kidney injury

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ABSTRACT

*Study objective:* To determine the appropriate mean arterial pressure (MAP) control level for elderly patients with hypertension during the perioperative period.
*Design:* A prospective, randomized study.
*Setting:* Three teaching hospitals in China.
*Patients:* Six hundred seventy-eight elderly patients with chronic hypertension undergoing major gastrointestinal surgery.
*Interventions:* Patients were randomly allocated to three groups and the target MAP level was strictly controlled to one of three levels: level I (65–79 mm Hg), level II (80–95 mm Hg), or level III (96–110 mm Hg).
*Measurements:* The primary outcome was acute kidney injury (AKI) (50% or 0.3 mg·dL$^{-1}$increase in creatinine level) during the first 7 postoperative days. The secondary outcomes were perioperative adverse complications. Moreover, vasoactive agents were observed during surgery.
*Main results:* The overall incidence of postoperative AKI was 10.9% (71/648). AKI occurred significantly less often in patients with level II MAP control (6.3%;13/206) than in patients with level I (13.5%; 31/230) and level III (12.9%; 27/210) ($P < 0.001$) MAP control. Level II was associated with lower incidences of hospital-acquired pneumonia (6.7%; 14/206; $P = 0.014$) and admission to the intensive care unit (ICU) (4.4%; 9/206; $P = 0.015$) and with shorter length of stay in the ICU ($P = 0.025$) when compared with level I and level III. Use of norepinephrine, phenylephrine, and nitroglycerin was significantly higher for patients with level III MAP control than for patients with level I and level II MAP control ($P = 0.001$).
*Conclusions:* For elderly hypertensive patients, controlling intraoperative MAP levels to 80 to 95 mm Hg can reduce postoperative AKI after major abdominal surgery.



## 1. Introduction

Acute kidney injury (AKI) is a significant clinical problem with a high rate of mortality and morbidity that affects 7.5% of patients who undergo noncardiac surgery [1–3]. A recent study showed that surgical patients with postoperative AKI are eight-times more likely to die within 30 days after surgery [4]. A large, retrospective study of 3.6 million veterans who underwent major surgery showed that patients with postoperative AKI had more negative outcomes than patients without AKI. For instance, patients with postoperative AKI often had longer hospitalizations, higher rates of 30-day hospital readmission, and higher 1-year mortality rates [5]. Many risk factors have been proposed to contribute to the occurrence of postoperative AKI, such as preexisting renal

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dysfunction, obesity, type of surgery, intra-abdominal pressure, and perioperative hemodynamic goals [6]. However, none of these proposed risk factors had been shown to be the key contribution to the occurrence of postoperative AKI.

Perioperative hypotension was recently proposed as an important determinant of postoperative AKI [5,7]. A large retrospective study revealed that the risk of postoperative AKI is significantly increased in surgical patients with >1 min of mean arterial pressure (MAP) lower than 55 mm Hg and >5 min of MAP from 55 to 59 mm Hg [2]. Another single-center cohort study demonstrated that postoperative AKI was associated with >10 min of intraoperative MAP lower than 55 mm Hg and 11–20 min of MAP lower than 60 mm Hg [8]. Asfar et al. [9] conducted a multicenter study involving 776 septic shock patients and showed that a target MAP of 65–70 mm Hg for patients without prior chronic hypertension and MAP of 80–85 mm Hg for patients with previous hypertension significantly lowered the incidence of postoperative AKI and the need for continuous renal replacement. These findings highlight the important role of MAP in postoperative AKI; however, the heterogeneity of the study subjects in these previous studies prevented understanding the appropriate MAP level during the perioperative period for elderly patients.

We performed a prospective, randomized study to determine the appropriate intraoperative MAP management level for elderly patients with chronic hypertension. The risks of three intraoperative MAP levels, 65–79 mm Hg, 80–95 mm Hg, and 96–110 mm Hg, for postoperative AKI were separately evaluated. Furthermore, we hypothesized that one of three intraoperative MAP levels might be suitable for elderly hypertensive patients and significantly reduce AKI after surgery.

## 2. Materials and methods

### 2.1. Study design and ethics

This was a prospective, randomized, and open-label study conducted at three teaching hospitals in China. This study was registered at www.Chictr.org.cn (ChiCTR-ROC-15006892) on August 7, 2015, and it was performed between August 24, 2015 and August 24, 2016. Eligible patients were randomly allocated to one of the following three groups: MAP, 65–79 mm Hg; MAP, 80–95 mm Hg; and MAP, 96–110 mm H. The study protocol was approved by the institutional ethics committees. Signed informed consents were obtained from all participants or their relative caregivers. This study was overseen by an independent data and safety monitoring group to ensure the safety of the participants, the validity of the data, and the credibility of the study results. Furthermore, investigators who collected follow-up information were blinded to the intervention status. All analyses were performed by an independent senior statistician before the randomization code was broken.

### 2.2. Subjects

We recruited patients who had chronic hypertension (diagnosed by systolic blood pressure > 140 mm Hg and/or diastolic blood pressure > 90 mm Hg in the absence of antihypertensive medications) and were scheduled for elective major gastrointestinal surgery (gastric cancer eradication surgery or colorectal cancer surgery) *via* either an open or a laparoscopic route. Patients were included in the study if all of the following criteria were met: 1) patients were 65–80 years old; 2) patients had American Society of Anesthesiologists (ASA) physical status grade I to III disease with a predicted surgery time >60 min; 3) no surgery for preexisting renal disease; 4) current left ventricular ejection fraction >50%; and 5) no sign of cardiac dysfunction. Patients were excluded from the study if any of the following were true: 1) patients used non-steroidal anti-inflammatory drugs during the past month; 2) patients had heart failure during the past 2 months; 3) patients had myocardial infarction during the past month (confirmed by blood-specific enzymes); 4) current severe pulmonary function insufficiency; 5) current intermediate to severe pulmonary hypertension; and 6) chronic kidney diseases or renal dysfunction (confirmed by previous physician's diagnosis). Detailed information regarding patients' adherence to the antihypertensive drug regimen or the adequacy of the antihypertensive treatment was obtained before recruitment.

### 2.3. Anesthesia protocol

All patients were intravenously injected with 1–3 mg midazolam 30 min before surgery. After entering the operation room, left radial artery catheterization guiding by Doppler ultrasound was performed under local anesthesia. The FloTrac/Vigileo system (MHD8; Edwards Lifesciences, Irvine, CA, USA) was used to obtain the cardiac output/cardiac index (CI), stroke volume (SV), stroke volume variation (SVV), and other hemodynamic parameters. A 16-G intravenous line was inserted into the right internal jugular vein under B-wave ultrasound guidance for fluid infusion and intermittent monitoring of central venous pressure (CVP).

The anesthesia induction agents were propofol (plasma concentration 4–5 $\mu g \cdot mL^{-1}$ under target controlled infusion), fentanyl (3–5 $\mu g \cdot kg^{-1}$), and *cis*-atracurium (0.15–0.2 $mg \cdot kg^{-1}$). These were maintained with continuous infusion of remifentanil (effect site concentration 6–8 $ng \cdot mL^{-1}$) and propofol (effect site concentration 3–4 $\mu g \cdot mL^{-1}$) by targeted controlled infusion. The depth of anesthesia was monitored by the bispectral index (Aspect Medical System, Saint Charles, USA) and its value was kept between 45 and 60. The *cis*-atracurium (0.004 $mg \cdot kg^{-1} \cdot min^{-1}$) was continuously infused to optimize muscle relaxation during surgery.

### 2.4. Fluid therapy

To ensure the appropriate volume status of the patients, a constant 7- to 8-$mL \cdot kg^{-1} \cdot h^{-1}$ crystalloid bolus fluid infusion was executed to maintain SVV at 8–13% and urine output at >1.0 $mL \cdot kg^{-1} \cdot h^{-1}$ during surgery (Fig. 1) [10]. Consecutive patients received an additional bolus of crystalloid 1.0 $mL \cdot kg^{-1}$ for each fasted hour from 8:00 AM until anesthesia induction. For patients undergoing laparoscopic surgery, the pneumoperitoneum insufflation pressure was set at 10–14 mm Hg. The FloTrac/Vigileo device was used to measure SVV and other hemodynamic parameters; 200 mL 6% hydroxyethyl starch was induced within 15 min each time, with SVV between 10% and 13%, and monitored by the FloTrac/Vigileo system. When the measured SVV was 13% more than the normal level (lasting for 5 min), or when the current subset reaction was positive (SV increased >10%), an additional 200 mL 6% hydroxyethyl starch was introduced. Blood transfusion was performed to control hemoglobin levels > 90 $g \cdot L^{-1}$ according to perioperative blood transfusion guidelines [11], and intraoperative blood gas analysis was tested every 30 min during surgery. The body temperatures of all patients were maintained at higher than 36 °C using an insulation blanket.

### 2.5. MAP control protocol

Vasoactive agents such as noradrenaline (0.03–0.3 $\mu g \cdot kg^{-1}$ $min^{-1}$), phenylephrine (10–100 μg each bolus), nitroglycerin (0.03–0.6 $\mu g \cdot kg^{-1} \cdot min^{-1}$), and phentolamine (0.5–3 mg every bolus) were introduced to adjust the MAP level. The initial dose for continuous infusion of noradrenaline or nitroglycerin was 0.03 $\mu g \cdot kg^{-1}$ $min^{-1}$. Noradrenaline and nitroglycerin were selected for continuous infusion, whereas phenylephrine and phentolamine were only used for bolus injections. If the current MAP deviated from the target goal, then it was corrected to the target level within 5 min using the aforementioned agents. If repeated bolus injections were used more than four times, and if the MAP level still could not be titrated to the target goal, then continuous infusion was initiated in increments or decrements of 0.03 $\mu g \cdot kg^{-1}$ $min^{-1}$ for at least 3 min. The vasoactive agents were discontinued when the current MAP level was returned to the target level for 5 min. When sudden fluctuation in perioperative blood pressure occurred, the following factors should be firstly considered, such as visceral stretch, vena cava compression, massive hemorrhage and other harmful stimuli. A 10% deviation in the MAP level from the target goal within 5 min was allowed. Fluid therapy and the MAP pressure control algorithm are shown in Fig. 1. Three intraoperative MAP levels were separately evaluated.

### 2.6. Study outcomes

The primary outcome was the incidence of AKI after major abdominal surgery. Postoperative serum creatinine level increases >50% or >0.3 mg·dL$^{-1}$ from baseline were regarded as AKI. AKI was diagnosed according to the criteria of the Kidney Disease: Improving Global Outcome (KDIGO) by considering the percentage of maximal increase in serum creatinine (△Cr) levels during the first 7 postoperative days (PODs): $\Delta Cr = \text{Maximum}\ (Cr_{POD1}, Cr_{POD2}, \ldots, Cr_{POD7}) - Cr_{Prep} / Cr_{Prep} \times 100\%$ [12,13]. Serum creatinine levels were routinely measured 1 day before surgery and 2, 3, and 7 days after surgery using the picric acid method. Renal replacement therapy was defined as any use of intermittent hemodialysis.

Secondary outcomes were the incidence of surgical site infection, hospital-acquired pneumonia, stroke, admission to the intensive care unit (ICU), stay in the ICU, length of hospital stay, and 28-day mortality. Postoperative complications were diagnosed based on the definitions of the International Surgical Outcome Study [14].

### 2.7. Sample size

The sample size was calculated according to prior studies that reported an AKI incidence of 11.8% for patients who underwent non-cardiac major surgery [5,15]. We hypothesized that the stringent MAP control could reduce the AKI incidence from 11.8% to 7.0% based on previous trials [1,2,8]. Consequently, enrollment of 210 patients in each group would obtain a power of 80% ($\beta = 0.2$) at a significance level of 0.05 ($\alpha = 0.05$, two-tailed). The dropouts, accounting for 5%, caused by withdrawal of consent or missing clinical data were compensated. Finally, recruitment of 221 cases in each group was determined.

### 2.8. Statistical analysis

SPSS software version 18.0 was used for data analysis. Normality of data distribution was assessed by Shapiro-Wilk test. Differences in patient characteristics and potential confounders among groups were compared using the one-way ANOVA for normally distributed continuous variables and the Kruskal-Wallis test for continuous variables that were not normally distributed. Comparisons between any two groups

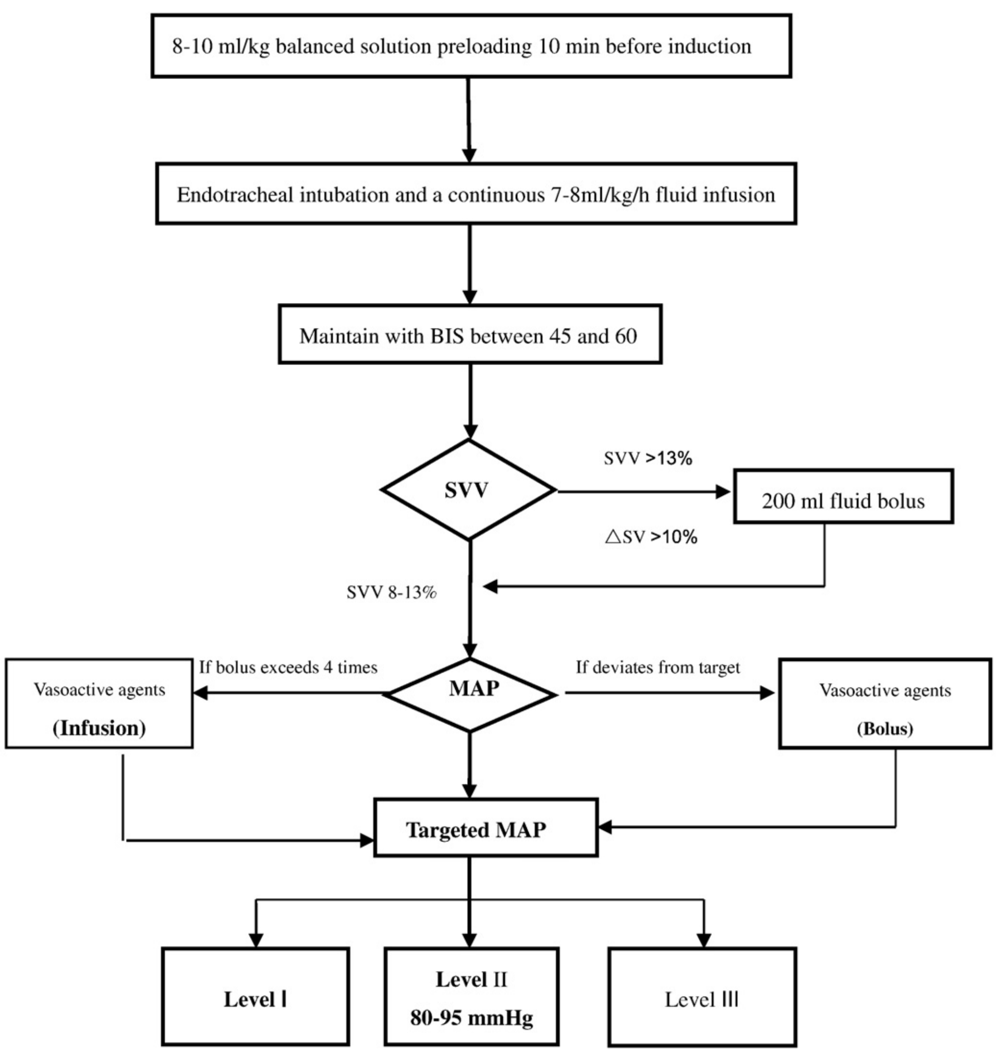


**Fig. 1.** The flow chart of fluid management and MAP control. BIS, bispectral index; SV, stroke volume; SVV, stroke volume variation; MAP, mean arterial pressure.

were corrected by the Bonferroni test. Dichotomous variables were compared using the Pearson's chi-square or Fisher's exact test when appropriate.

## 3. Results

### 3.1. Study population

1230 patients were screened at three teaching hospitals from August 24, 2015 to August 24, 2016. A total of 552 patients were excluded (Fig. 2). Finally, 678 patients were randomly allocated (1:1:1) to three MAP levels, namely, level I (65 to 79 mm Hg), level II (80 to 95 mm Hg), and level III (96 to 110 mm Hg), and 646 patients with complete data sets were included in the final analysis (level I = 230, level II = 206, level III = 210) (Fig. 2). All patients were older than 65 years, and the demographics and baseline inpatient characteristics among the three groups were similar (Table 1).

### 3.2. Intraoperative data and management

Intraoperative data were not significantly different among the three groups, including duration of anesthesia and surgery, volume of intravenous fluids, amount of plasma and red blood cells, blood loss, and intraoperative urine output (Table 2). Patients with MAP level III were administered larger doses of norepinephrine (3.9 ± 1.0 mg), phenylephrine (700 ± 202 μg), and nitroglycerin (5.4 ± 1.6 mg) compared with patients with MAP level I and level II ($P < 0.01$) (Table 2). The time weighted average-mean arterial pressure (TWA-MAP) for the three MAP control levels (I, II and level III) were 72 ± 5 mm Hg, 88 ± 7 mm Hg and 100 ± 6 mm Hg respectively, which implied that the actual MAP control level were within the preset targeted MAP level. And

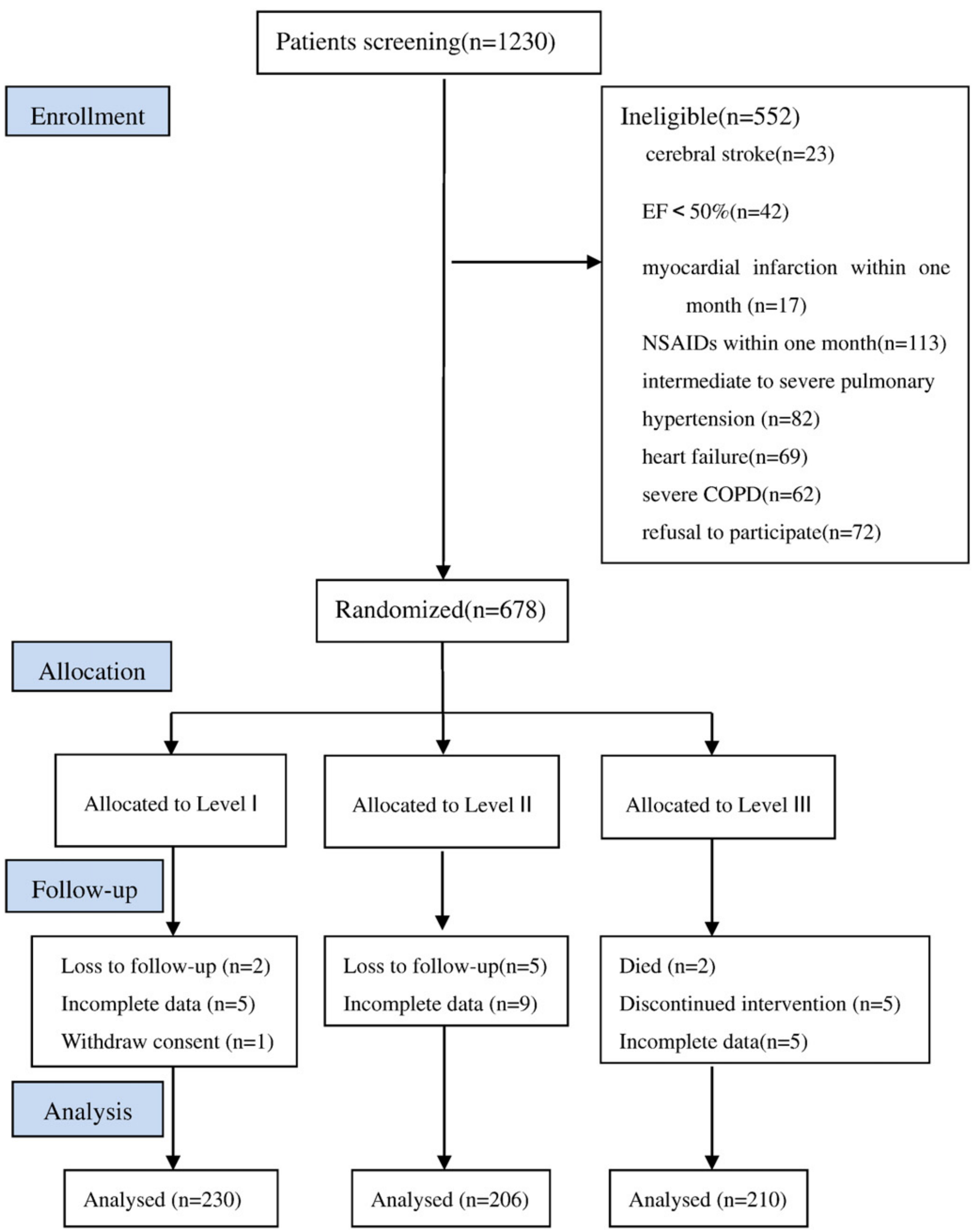


**Fig. 2.** CONSORT flow chart of the study.

**Table 1**
Demographic data and baseline characteristics of all patients.

| | I (n = 230) | II (n = 206) | III (n = 210) |
|---|---|---|---|
| Age (years) | 73 ± 7 | 73 ± 6 | 74 ± 5 |
| Gender (male) | 124 (60.2%) | 157 (68.3%) | 143 (65.6%) |
| Body weight (kg) | 69.8 ± 4.7 | 69.9 ± 5.1 | 70.1 ± 5.0 |
| ASA grade | | | |
| I | 28 (12.2%) | 28 (13.6%) | 27 (12.9%) |
| II | 169 (73.8%) | 140 (67.9%) | 159 (75.7%) |
| III | 32 (14%) | 38 (14.5%) | 24 (11.4%) |
| NYHA grade | | | |
| I | 26 (10.7%) | 26 (12.6%) | 22 (10.5%) |
| II | 189 (83.7%) | 165 (80.1%) | 174 (82.9%) |
| III | 15 (5.6%) | 15 (7.3%) | 14 (6.7%) |
| Past history | | | |
| Smoke | 102 (44.3%) | 98 (47.5%) | 95 (45.2%) |
| Alcohol | 87 (37.8%) | 80 (38.8%) | 83 (39.5%) |
| Stroke | 8 (3.5%) | 6 (2.9%) | 9 (4.28%) |
| TIA | 11 (4.78%) | 13 (6.31%) | 7 (3.59%) |
| COPD | 4 (1.74%) | 3 (1.45%) | 4 (1.9%) |
| Diabetes mellitus | 56 (24.3%) | 48 (23.3%) | 52 (24.7%) |
| Hyperthyroidism | 3 (0.9%) | 2 (0.9%) | 4 (1.2%) |
| Antihypertensive agents | | | |
| ACEI | 54 (23.5%) | 48 (23.3%) | 52 (24.8%) |
| ARB | 90(39.1%) | 82(39.8%) | 87 (41.4%) |
| β-Blockers | 60 (26.1%) | 53 (25.7%) | 58 (27.6%) |
| CCB | 42 (18.3%) | 36 (17.5%) | 40 (19.0%) |
| Diuretics | 30(13.0%) | 27(13.1%) | 28(13.3%) |
| Others | 34 (14.8%) | 39 (17.4%) | 33 (15.7%) |
| Hb (g/L) | 123.7 ± 22.0 | 124.9 ± 21.6 | 120.6 ± 17.9 |
| WBC ($\times 10^9$/L) | 5.8 ± 1.7 | 6.3 ± 2.2 | 5.5 ± 2.0 |
| BUN (mmol/L) | 4.95 ± 1.65 | 4.88 ± 1.66 | 5.25 ± 1.65 |
| SCr (mmol/L) | 54.3 ± 7.9 | 52.9 ± 5.7 | 53.8 ± 6.7 |
| FBS (mmol/L) | 5.5 ± 1.5 | 5.5 ± 1.4 | 5.3 ± 1.7 |
| EF (%) | 65 ± 4 | 66 ± 5 | 64 ± 6 |
| Baseline BP | | | |
| SBP (mm Hg) | 146 ± 20 | 150 ± 18 | 145 ± 17 |
| DBP (mm Hg) | 84 ± 10 | 86 ± 11 | 82 ± 9 |
| MAP (mm Hg) | 102 ± 23 | 104 ± 22 | 103 ± 21 |

Data are expressed as mean (SD) or number (%).
Baseline BP is calculated as the average of all radial cuff pressure 2 to 3 days (at least 3 times) before surgery in the ward. Definitions of smoke and alcohol were based on WHO issued in 1997. The diagnosis of COPD in terms of international GOLD guideline published in 2015. ASA, American Association of Anesthesiologists; NYHA, New York Heart Association; TIA, transient ischemic attack; COPD, chronic obstructive pulmonary disease; ACEI, angiotensin converting enzyme inhibitor; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; WBC, white blood cell; Hb, hemoglobin; BUN, blood urine nitrogen; SCr, serum creatinine; BP, blood pressure; FBS, fasting blood sugar; EF, ejection fraction; SBP, systolic blood pressure; DBP, diastolic blood pressure; MAP, mean arterial pressure.

**Table 2**
Intraoperative data among three groups.

| | I (n = 230) | II (n = 206) | III (n = 210) | *P* value |
|---|---|---|---|---|
| Anesthetic time (min) | 220.6 ± 71.0 | 212.9 ± 73.6 | 218.9 ± 69.2 | 0.89 |
| Surgical time (h) | | | | |
| ≦2 | 110 (47.8%) | 100 (48.5%) | 103 (49.0%) | 0.67 |
| 2–4 | 105 (45.7%) | 94 (45.6%) | 94 (44.8%) | |
| ≧4 | 15 (6.5%) | 12 (5.9%) | 13 (6.2%) | |
| Surgical route | | | | |
| Open | 102 (44.3%) | 94 (45.6%) | 100 (47.6%) | |
| Laparoscopy | 108 (46.9%) | 96 (46.6%) | 93 (44.3%) | 0.35 |
| Laparoscopy to open | 20 (8.8%) | 16 (7.8%) | 17 (8.1%) | |
| Fluid management | | | | |
| Crystalloids (mL) | 2102 ± 632 | 2153 ± 707 | 2260 ± 649 | 0.13 |
| Colloids (mL) | 756 ± 350 | 698 ± 332 | 715 ± 305 | 0.36 |
| Plasma (mL) | 320 ± 160 | 310 ± 150 | 325 ± 162 | 0.08 |
| RBC (mL) | 150 ± 150 | 170 ± 120 | 160 ± 150 | 0.22 |
| Estimate blood loss (mL) | | | | |
| ≦100 | 110 (47.8%) | 102 (49.5%) | 112 (53.3%) | 0.56 |
| 101–500 | 112 (48.7%) | 96 (46.5%) | 88 (42.0%) | |
| 501–800 | 5 (2.2%) | 6 (3.0%) | 7 (3.3%) | |
| 801–1000 | 3 (1.3%) | 2 (1.0%) | 3 (1.4%) | |
| Urine output (mL) | | | | |
| ≦500 | 30 (13.0%) | 26 (12.6%) | 28 (13.3%) | 0.21 |
| 501–1000 | 188 (81.7%) | 169 (82.0%) | 172 (81.9%) | |
| >1000 | 12 (5.3%) | 9 (4.4%) | 10 (4.8%) | |
| Vasoactive agents | | | | |
| Norepinephrine (mg) | 2.2 ± 1.2 | 2.1 ± 1.5 | 3.9 ± 1.0 | 0.001 |
| Phenylephrine (μg) | 450 ± 155 | 507 ± 165 | 700 ± 202 | 0.001 |
| Nitroglycerin (mg) | 3.1 ± 1.2 | 3.3 ± 1.5 | 5.4 ± 1.6 | 0.001 |
| Phentolamine (mg) | 6.6 ± 4.2 | 7.3 ± 3.8 | 7.0 ± 3.5 | 0.14 |
| Esmolol (mg) | 65 ± 32 | 60 ± 40 | 69 ± 38 | 0.35 |
| Atropine (mg) | 0.8 ± 0.4 | 0.9 ± 0.4 | 0.8 ± 0.4 | 0.09 |
| TWA-MAP (mm Hg) | 72 ± 5 | 88 ± 7 | 100 ± 6 | 0.001 |

Data are expressed as mean (SD) or number (%).
RBC, red blood cell; TWA-MAP, time weighted average-mean arterial pressure.
TWA-MAP is calculated as the MAP measurements divided by total measurement time (all measurements are equidistant of 1 min interval since we placed arterial line in every case and easy to extract MAP data from the electronic record system) [16].

the actual MAP control level was the essential determinant of success in this study.

*3.2.1. Perioperative occurrence of AKI and other adverse outcomes*

AKI was observed for 10.9% (71/646) of patients after surgery. Patients with MAP level II exhibited a lower rate of AKI (6.3%) compared to the patients with MAP level I (13.5%) and level III (12.9%) (Table 3). When divided according to AKI stage, the numbers of patients with KDIGO stage 1 or KDIGO stage 2 were also significantly different among the three groups ($P < 0.01$). We did not find any patients with KDIGO stage 3 among the three groups. The incidence of ICU admission was significantly lower for patients with MAP level II (4.4%) than for patients with MAP level I (8.4%) and level III (7.6%; $P = 0.015$). There were no significant differences in 28-day mortality among the three groups (level I, 3.0%; level II, 2.9%; level III, 3.8%) (Table 3).

## 4. Discussion

In this prospective and randomized study, we characterized the incidence of postoperative AKI and other clinical complications in elderly patients with chronic hypertension who underwent major gastrointestinal surgery at three rigorously controlled intraoperative MAP levels. It was revealed that intraoperative MAP controlled at 80–95 mm Hg can significantly reduce the incidence of postoperative AKI and other postoperative complications in elderly patients with chronic hypertension.

Our reported results as regards to postoperative AKI occurrence are consistent with previous study [5]. The overall incidence of postoperative AKI was 10.9% in the current study, however, it was 6.3% for patients for MAP control of 80-95 mm Hg, which is significantly lower than that for patients with MAP control of 65–79 mm Hg (13.5%) and 96–110 mm Hg (12.9%). In fact, there is a broad range

**Table 3**
Perioperative occurrence of AKI, and other adverse outcomes.

| | I (n = 230) | II (n = 206) | III (n = 210) | *P* value |
|---|---|---|---|---|
| Primary outcome | | | | |
| Incidence of AKI n(%) | 31 (13.5%) | 13 (6.3%) | 27 (12.9%) | 0.033 |
| KDIGO stage1 n(%) | 20 (8.7%) | 9 (4.4%) | 20 (9.5%) | |
| KDIGO stage2 n(%) | 11 (4.5%) | 4 (1.9%) | 7 (3.4%) | |
| KDIGO stage3 n(%) | 0 (−) | 0 (−) | 0 (−) | |
| Secondary outcome | | | | |
| Surgical site infection n(%) | 9 (3.9%) | 7 (3.4%) | 7 (3.4%) | 0.542 |
| Hospital acquired pneumonia n(%) | 26 (11.3%) | 14 (6.7%) | 22 (10.4%) | 0.014 |
| Stroke n(%) | 1 (0.43%) | 1 (0.48%) | 1 (0.47%) | 0.341 |
| Admission to ICU n(%) | 19 (8.4%) | 9 (4.4%) | 16 (7.6%) | 0.015 |
| Stay in ICU d(IQR) | 2 (1–7) | 1 (1–3) | 2 (1–6) | 0.025 |
| Mortality of 28 day n(%) | 7 (3.0%) | 6 (2.9%) | 8 (3.8%) | 0.671 |

Data are expressed as mean (SD) or number (%).
ICU = intensive care unit; IQR = interquartile range.
KDIGO stage 1 = △Cr 50% to 99%; KDIGO stage 2 = △Cr 100% to 199%; KDIGO stage 3 = △Cr 200% to 299% or serum creatinine above 353.6 μmol/L or initiate renal replacement therapy.

of incidence (1.0% to 7.5%) of postoperative AKI for noncardiac surgery patients owing to the heterogeneity of study subjects, criteria for AKI diagnosis and stage, type of surgery, comorbidity, preoperative renal function status, and others factors [17–19]. In this study, we enrolled elderly patients from 65 to 80 years and also focused on elderly patients with chronic hypertension, which might be the reason for the high incidence of postoperative AKI in our study.

A novel aspect of our study was that we found a middle MAP level (80–95 mm Hg), but not a low (65–79 mm Hg) or high (96–110 mm Hg) MAP level, can decrease the incidence of AKI and other postoperative complications. Although the exact mechanism for this phenomenon remains elusive, we postulated that a middle MAP level might be appropriate for the majority of elderly patients with hypertension, below or above this level will cause abnormal perfusion renal tissue, which involving autoregulation mechanism and other factors in organ preserved process, further decreases glomerular filtration rate and eventually jeopardizes kidney function based on the following literature [2,8,20–22]. Intuitively, abruptly fluctuating MAP, even briefly, can be deleterious and can lead to increased postoperative cardiovascular complications and mortality [23,24]. In our study, a strict MAP control protocol was used with a view to reducing the impact of MAP variation (5 min) on AKI, of which may be the one of the factors contributing the decreased occurrence of AKI in middle MAP level. Another large retrospective study demonstrated that intraoperative MAP < 60 mm Hg significantly increased the risk of AKI [25], which means that MAP level I inevitably causes insufficient renal perfusion in elderly patients with hypertension. Additionally, too high blood pressure is also deleterious to vital organ [26]. Therefore, we conclude that a MAP level between 80 and 95 mm Hg is the appropriate pressure for the low incidence of AKI.

As shown in Table 3, the rate of hospital-acquired pneumonia was higher for level I and level III; furthermore, lengths of stay in the ICU for level I and level III were longer than that for level II. The exact reason for the higher occurrence of pneumonia remains undetermined in this trial, partially because patients with level I and level III had longer ICU stays. During the whole observation period, patients with level II had a lower incidence of admission to the ICU. This could not be simply attributed to the contribution of the MAP level because the causes for entering the ICU are multifactorial. Unfortunately, we only recorded the duration of stay in the ICU and neglected the exact reason for admitting or readmitting to the ICU in this study.

The following studies, such as saline [27], chloride-restricted fluid [28], colloid solutions [29] and fluid balance [30], suggested that both the type and the amount of fluids are thought to affect the incidence of AKI. In our study, crystalloids and colloids were used to volume expansion, and real-time monitoring of SVV was used to evaluate volume status and make rapid adjustments of the volume to eliminate the effect of volume status on AKI. No significant difference in the amount of crystalloid and colloid solutions was observed for the three MAP control goals, suggesting that the significant differences in the incidence of AKI and other complications were not caused by fluid therapy.

In the current trial, level III required larger doses of vasopressors (norepinephrine and phenylephrine) and also larger doses of vasodilators like nitroglycerin. The reason for this phenomenon is that at different stages of surgery, the intensity of noxious stimulation varied, furthermore, under the same depth of anesthesia, a higher MAP level requires larger doses of vasoactives. Whether the larger doses of vasopressors aggravate renal function and further promote the occurrence of AKI in level III remains to be unknown. In septic patients, vasopressor may have a beneficial effect on renal function [31], whereas, administration of vasopressor was associated with renal vasculature constriction and renal tissue hypoperfusion which may compromise kidney function in non-septic patients [32].

There are several shortcomings in this study. First, the randomization *per se* created the possibility of exposing them to renal injury. Because it was a randomized study, we could not allocate patients according to preoperative baseline blood pressure, and some patients were randomized to the low MAP levels, which might not have been suitable for pre-determined MAP levels. Second, renal blood flow using ultrasound-tagged technology and cerebral oxygenation using the near-infrared spectroscopy technique under different MAP levels have not been monitored; therefore, the best suitable MAP level for the kidney or brain remains to be further solved. Third, the MAP control status after surgery, blood management, and antibiotic selection were left to the discretion of the attending surgeon; it was unknown whether these factors had effects on AKI. Finally, in the study, we only observed 28-day postoperative mortality and solely concentrated on a specific population of elderly patients with hypertension. Therefore, the results should be cautiously extrapolated to other patients.

In conclusion, in elderly patients with chronic hypertension undergoing major gastrointestinal surgery, a MAP level ranging from 80 to 95 mm Hg confers a protective role in the renal function, reduces postoperative AKI after major gastrointestinal surgery, and decreases the likelihood of other complications.

## Clinical trial registration

Registry URL: http://www.Chictr.org.cn. Clinical trial number: ChiCTR-ROC-15006892.

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